

Water bears

Phylum Tardigrada

Species About 1,000

Occurrence Freshwater, saltwater, wetlands

Similar to velvet worms, but microscopic, water bears have plump bodies and four pairs of short, fleshy legs, on which they crawl around. In damp terrestrial habitats, they feed on plant juices and other animals. Water bears have a remarkable tolerance to

drought, being able to shrivel and survive in a state of dehydration for several years, then swelling up and resuming normal life

when conditions improve. By being dormant for extended periods, they may live over 50 years.



Moss water bear *Echiniscus* species

Bryozoans

Phylum Bryozoa

Species About 6,000

Occurrence Attached to hard surfaces in aquatic environments

Also known as moss animals, bryozoans are very small aquatic animals that live in colonies. Each animal occupies a boxlike case that forms part of a mat or ribbon of identical cases; typically, these encrust rocks, seaweed, or flooded tree roots, but some form branched, plantlike structures. The animals feed by extending crowns of feathery tentacles to filter food from the water.

Ribbon worms

Phylum Nemertea

Species About 1,400

Occurrence On bed, in surface or middle waters of seas, rivers and lakes; forests

Related to flatworms, these mainly marine worms are armed with a muscular proboscis that normally lies inside the animal's snout, but can be pushed out by hydraulic pressure to catch prey; in some forms, it is armed with venomous barbs to inject toxins. Most ribbon worms are less than 8 in (20 cm) long, but one species is known to grow to 177 ft (54 m), making it the longest of all known animals. Many ribbon worms are brightly colored.

Brachiopods

Phylum Brachiopoda

Species About 400

Occurrence Attached to hard surfaces or buried in sediments on the seabed

Enclosed in bivalved shells, these marine animals look like bivalve mollusks, but they are not related and their anatomy is quite different. A brachiopod lives attached to a rock or burrow by a muscular stalk, and its shells protect a loop of hollow, fringed tentacles, called the lophophore, which filters food from the water. Also known as lamp shells, brachiopods were once far more abundant than they are now.

Spoonworms

Phylum Echiura

Species About 200

Occurrence In burrows or other cavities on the seabed

The echiurans, or spoonworms, are soft-bodied marine animals that occur in seabeds throughout the world. Although some species inhabit rock crevices, most live in burrows in the mud. A few species even occur in brackish water. Spoonworms have bulbous, sausagelike bodies and a spoon-shaped, nonretractable, muscular proboscis, which they use to gather food from surrounding rocks and sediments. Although usually only

about 6 in (15 cm) long, some can extend the proboscis for 39 in (100 cm) or more.



Green spoonworm *Bonellia viridis*



Arrow worms

Phylum Chaetognatha

Species About 150

Occurrence In plankton, one genus on the seabed

Most of these small, torpedo-shaped animals drift in the oceans among the plankton, where they attack and eat other animals. Arrow worms are armed with a fearsome array of grasping spines around the mouth that help them to crush their prey. The spines are backed up by piercing teeth that inject

a potent, paralyzing venom. Produced by symbiotic bacteria, this venom is called tetrodotoxin and is used by very few animals. Much of the prey of arrow worms consists of larval fish, which they devour in large numbers. Highly active predators, arrow worms can eat a third of their own body weight each day. Their size ranges from about 1/8 in (3 mm) to more than 4 in (100 mm); species inhabiting colder waters are generally larger than those from tropical seas. Widespread in open waters of oceans worldwide, their large numbers can have a significant ecological impact—for example, on fish larvae.

Arrow worm *Sagitta* species

Acorn worms

Phylum Hemichordata

Species About 130

Occurrence In marine mud and sand, and other intertidal and subtidal habitats

Acorn worms are named after the pointed front ends of their bodies, which are separated from the rest of the body by a tubelike collar. Typical acorn worms live in mucus-lined burrows in soft seabeds and tidal flats. They consume sand and mud, ejecting anything inedible as a coiled cast. However, others feed by extending a proboscis from the burrow and trapping small drifting organisms in sticky mucus.